QML Journal Club

Choong Pak Shen

March 27, 2026

QML Journal Club 1st meeting

Motivation

Quantum machine learning looks at the synergies between machine learning and quantum information, or how quantum computing may assist machine learning due to its different computing architecture.

Are quantum computers better at machine learning than classical computers? It depends.

1. Structure of data
2. Supervised machine learning or unsupervised machine learning
3. Quantum computing as a subroutine or optimizer
4. Types of quantum computer (annealer, superconducting qubit, continuous-variable systems)
5. Noise

What is quantum computing?

Quantum computer is a computing architecture that makes use of quantum physics to manipulate information.

The first quantum revolution happened around 20th-century, where classical physics was shown to be inadequate to explain several physical phenomena, such as blackbody radiation, photoelectric effect, and wave-particle duality.

The second quantum revolution started around 2000, where we have more matured technologies to control microscopic systems in a more precised manner, such that various quantum effects may be used in information processing.

These quantum systems are fragile and noisy. We are currently in the noisy-intermediate scale quantum (NISQ) era and moving towards fault-tolerant quantum era.

Characteristics of NISQ include:

1. Around 10-100 quantum bits (qubits)
2. No error correction

There are two types of algorithms,

1. Quantum algorithms: Algorithms that run on real quantum computers and produce theoretically proven quantum advantage. Current quantum hardware cannot implement these algorithms effectively.
2. NISQ algorithms: Algorithms that run on the current generation of quantum computers.

Quantum advantage

The holy grail of "quantum advantage"

1. Slower growth or asymptotic runtime with the size of dataset compared with a known machine learning algorithm?
2. Provable speedup?
3. Benchmarking against the best known classical algorithm, or computational complexity analysis?
4. Energy consumption?
5. The precision of output?
6. Probability of computational failure?
7. How much data to train a model?
8. How well can the model generalizes to different situations?

Claims of exponentially fast quantum machine learning algorithms come with a lot of assumptions.

Quantum machine learning

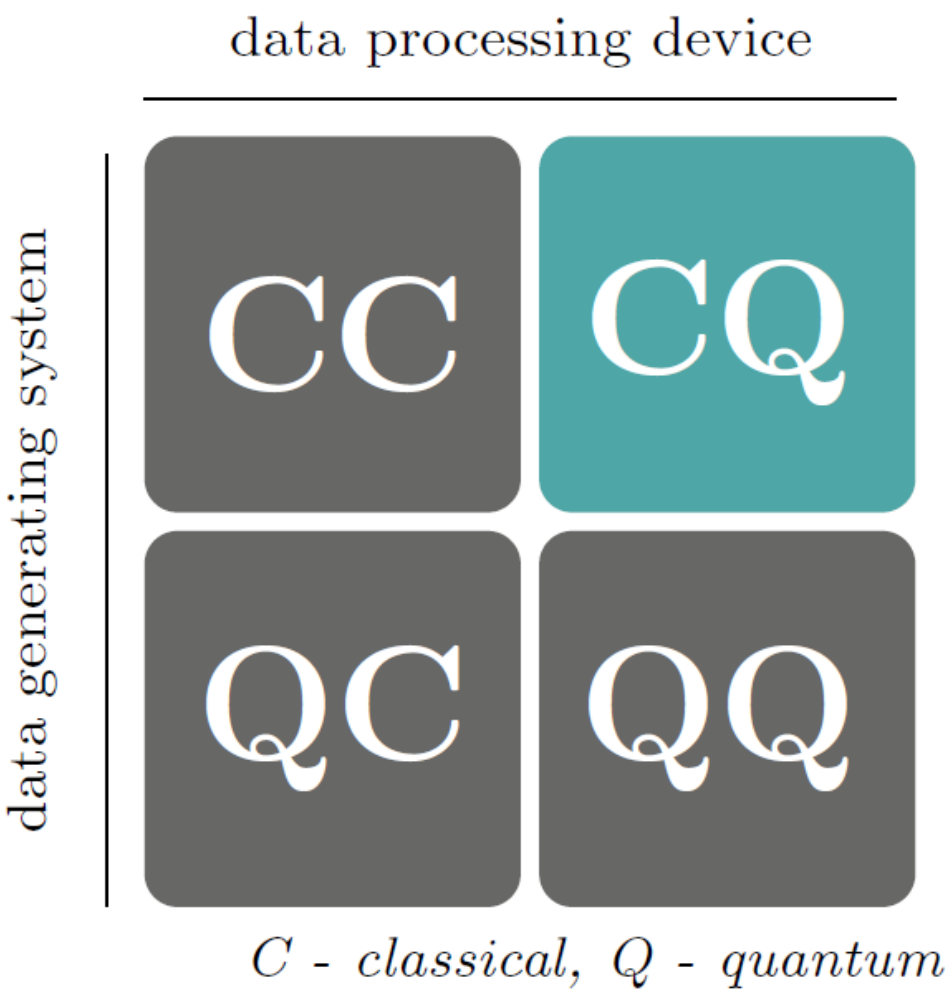


Figure 1: Four approaches to combine quantum computing and machine learning.

CC refers to classical data being processed classically. Examples would be the application of tensor networks, which was originally developed for quantum many-body systems to neural network training, or quantum-inspired machine learning models.

QC refers to quantum data being processed classically. Examples would be to use neural networks to discriminate quantum states or phase transitions in quantum many-body systems.

CQ refers to classical data being processed using quantum devices, and it is the scope of quantum machine learning. Examples would be to use time series data, satellite images or numerical data on quantum computers for analysis. Note that the data has to be loaded onto a quantum computer, which will be proportional to the data size.

QQ refers to quantum data being processed using quantum devices. Data could be derived from measuring a quantum system in an experiment, and then being fed back into a quantum computer, or quantum states. For an exponential speedup, quantum computer can access to quantum states (as data) directly.

Toy example: Nearest-neighbor classifier

Suppose you have two data points from the passenger list of Titanic, consisting of the price of their ticket (0 to 10,000), their cabin number (1 to 2500), and their survival. Given the same data point of a third passenger, can we identify the probability of survival for this third passenger?

	Raw data		Preprocessed data		
	Price	Cabin	Price	Cabin	Survival (y)
Passenger 1	8,500	0910	0.85	0.36	1 (yes)
Passenger 2	1,200	2105	0.12	0.84	0 (No)
Passenger 3	7,800	1121	0.78	0.45	?

The nearest-neighbor (squared-distance) classifier defines a closeness measure between the data points and the new input,

$$p(y=1|\vec{x}) = \frac{1}{\chi}\frac{1}{M}\sum_{m|y=1}\left(1-\frac{1}{c}\left\|\vec{x}-\vec{x}_m\right\|^2\right), \tag{1}$$

and it is based on the assumption that similar inputs should have a similar output.

Quantum squared-distance classifier

Step 1: Normalizing the length of each input vector to 1.

Passenger 1: $0.85^2 + 0.36^2 = 0.8521$, $\sqrt{\frac{0.85^2}{0.8521}} = 0.921$, $\sqrt{\frac{0.36^2}{0.8521}} = 0.39$

Passenger 2: $0.12^2 + 0.84^2 = 0.72$, $\sqrt{\frac{0.12^2}{0.72}} = 0.141$, $\sqrt{\frac{0.84^2}{0.72}} = 0.99$

Passenger 3: $0.78^2 + 0.45^2 = 0.8109$, $\sqrt{\frac{0.78^2}{0.8109}} = 0.866$, $\sqrt{\frac{0.45^2}{0.8109}} = 0.50$

Step 2: Data encoding by amplitude encoding.

Number of features: 6

Number of class labels: 2

Initial state is given by 4 qubits,

$$
\begin{aligned}
|\psi\rangle = \frac{1}{\sqrt{4}}([0.921|000\rangle + 0.39|001\rangle] \otimes |1\rangle + [0.141|010\rangle + 0.99|011\rangle] \otimes |0\rangle \\
+ [0.866|100\rangle + 0.50|101\rangle] \otimes |1\rangle + [0.866|110\rangle + 0.50|111\rangle] \otimes |0\rangle) \\
= \frac{1}{\sqrt{4}}(0.921|0001\rangle + 0.39|0011\rangle + 0.141|0100\rangle + 0.99|0110\rangle \\
+ 0.866|1001\rangle + 0.50|1011\rangle + 0.866|1100\rangle + 0.50|1110\rangle)
\end{aligned}
$$

Step 3: Apply Hadamard gate on the first qubit.

$$
\begin{aligned}
|\psi'\rangle = H_1|\psi\rangle = \frac{1}{\sqrt{8}}([0.921 + 0.866]|0001\rangle + [0.39 + 0.5]|0011\rangle \\
+ [0.141 + 0.866]|0100\rangle + [0.99 + 0.5]|0110\rangle \\
+ [0.921 - 0.866]|1001\rangle + [0.39 - 0.50]|1011\rangle \\
+ [0.141 - 0.866]|1100\rangle + [0.99 - 0.50]|1110\rangle)
\end{aligned}
$$

Step 4: Measure the first qubit and select samples (post-selection process) with $|q_1\rangle = |0\rangle$.

$$
\begin{aligned}
|\psi''\rangle = (M_1 + \mathrm{PS})|\psi'\rangle = \frac{1}{\sqrt{8\chi}}([0.921 + 0.866]|0001\rangle + [0.39 + 0.5]|0011\rangle \\
+ [0.141 + 0.866]|0100\rangle + [0.99 + 0.5]|0110\rangle)
\end{aligned}
$$

where χ is a normalization constant,

$$
\begin{aligned}
\chi &= \frac{1}{8}(|0.921 + 0.866|^2 + |0.39 + 0.5|^2 + |0.141 + 0.866|^2 + |0.99 + 0.5|^2) \\
&= 0.902
\end{aligned}
$$

Step 5: Measure the last qubit.

$$p(|q_4\rangle = |0\rangle) = 0.448$$
$$p(|q_4\rangle = |1\rangle) = 0.552$$

Alternatively, Step 4 and 5 can be combined by calculating the conditional probability,

$$\begin{aligned}
&p(|q_4\rangle = |0\rangle || q_1\rangle = |0\rangle) \\
&\quad = \frac{p(|q_4\rangle = |0\rangle) \cap p(|q_1\rangle = |0\rangle)}{p(|q_1\rangle = |0\rangle)} \\
&\quad = \frac{(0.141 + 0.866)^2 + (0.99 + 0.5)^2}{(0.921 + 0.866)^2 + (0.39 + 0.5)^2 + (0.141 + 0.866)^2 + (0.99 + 0.5)^2} \\
&\quad = 0.448
\end{aligned}$$

Let

$$a^2 + b^2 = 0.141^2 + 0.99^2 = 1,$$
$$c^2 + d^2 = 0.866^2 + 0.5^2 = 1.$$

We observe that

$$\begin{aligned}
p(|q_4\rangle = |0\rangle) &= \frac{1}{8\chi}[(0.141 + 0.866)^2 + (0.99 + 0.5)^2] \\
&= \frac{1}{8\chi}[(a + c)^2 + (b + d)^2] \\
&= \frac{1}{8\chi}[a + 2ac + c^2 + b^2 + 2bd + d^2] \\
&= \frac{1}{8\chi}[2 + 2ac + 2bd] = \frac{1}{4\chi}[1 + 1ac + 1bd] \\
&= \frac{1}{4\chi}[2 - (1 - ac - bd)] \\
&= \frac{1}{4\chi}[2 - \frac{1}{2}(a^2 + b^2 + c^2 + d^2 - 2ac - 2bd)]
\end{aligned}$$

$$
\begin{aligned}
p(|q_4\rangle = |0\rangle) &= \frac{1}{4\chi}[2 - \frac{1}{2}(\{a-c\}^2 + \{b-d\}^2)] \\
&= \frac{1}{2\chi}[1 - \frac{1}{4}(\{a-c\}^2 + \{b-d\}^2)] \\
&= \frac{1}{2\chi}[1 - \frac{1}{4}([0.141 - 0.866]^2 + [0.99 - 0.5]^2)]
\end{aligned}
$$

We see that this is the same as

$$
p(y=1|\vec{x}) = \frac{1}{\chi}\frac{1}{M}\sum_{m|y=1}\left(1 - \frac{1}{c}\left\|\vec{x} - \vec{x}_m\right\|^2\right).
$$

Insights from the toy example

The mathematical formalism of quantum theory provides an equivalent counterpart to machine learning. To some extent, quantum machine learning algorithms are often inspired by classical machine learning algorithms, and vice versa.

Data encoding is the most crucial step of quantum machine learning with classical data (CQ).

Quantum machine learning algorithms often requires further preprocessing on classical data.

The result of a quantum machine learning algorithm always require measurement.

QML Journal Club 2nd meeting

What is machine learning?

Artificial intelligence (AI) are computer systems that are good at human-like tasks, such as reasoning, navigating, and interacting.

Machine learning are computer algorithms that specialize in pattern recognition, i.e. generalize from patterns found in data to make predictions in unknown situations.

The patterns to learn are usually very complex and we have little understanding of the system's dynamics.

It is difficult to hand-shape a model of dynamic equations that captures the mechanism producing the data.

Hence, machine learning starts with a very general, agnostic mathematical model, and uses data to adapt it to the ease, a process called "training".

We do not necessarily gain information on the physical mechanism or statistical relationships behind the data, but consider it as a black box that has learned to produce reliable outputs.

The biggest criticism against machine learning is that its tools are opaque and make decisions that are difficult to verify.

This is also the biggest advantage of machine learning, because they can be generically applied to a huge variety of tasks and across different domains.

Explainable AI (XAI), interpretable AI or explainable machine learning (XML), explores methods that allow humans to understand the reasoning behind the decisions or predictions made by the algorithms.

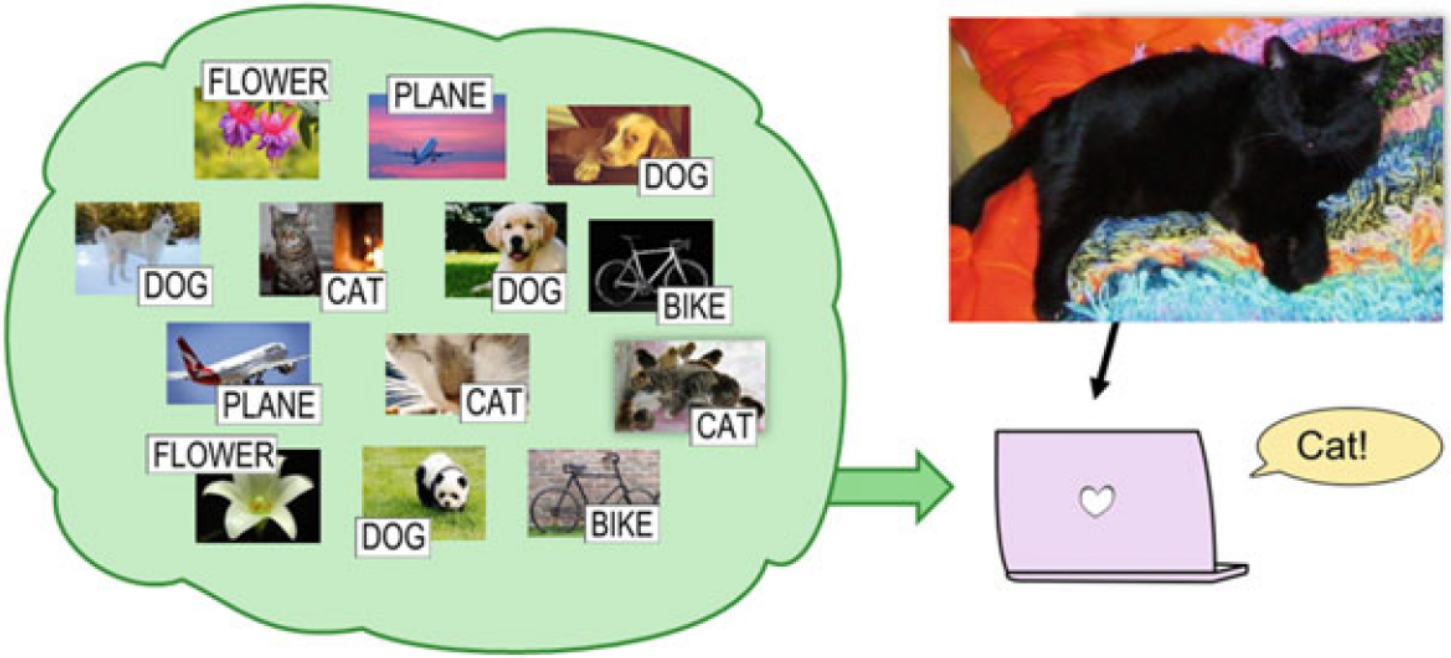
Historically, Kepler's laws of planetary motion were discovered by the accurate astronomical data of Tycho Brahe. Later, Newton's law of universal gravitation was formulated.

Types of machine learning

Supervised learning can be identified with labeled training data consisting of input-output pairs.

Example 1: Image recognition

In machine learning, one presents a large number of already labeled images to the computer from which it is supposed to learn the relationship of the digital image representation and its content.



Example 2: Time series forecasting

A time series is a set of data points recorded in consecutive time intervals. These data points will likely be correlated to many independent variables that are not recorded. A machine learning task could be used to predict what is going to happen for the next time step.

Unsupervised learning can be identified with data that does not come with target outputs. The model identifies patterns or structures in unlabeled data.

Example 3: Hypothesis guessing

A candidate is given a list of integers between 1 and 100, for example 4, 16, 36, and 100, and has to find new instances produced by the same rule. To do so, he has to guess the rule or hypothesis with which these numbers were randomly generated. One intuitive way of judging different hypotheses that all fit the data is to prefer those that have a smaller amount of options, since it would be a much bigger coincidence to pick data with a larger set. In probabilistic terms, one may prefer the hypothesis for which generating exactly the given dataset has the highest probability.

Hypothesis 1:
Even numbers

2	4	6	8	10
12	14	16	18	20
22	24	26	28	30
32	34	36	38	40
42	44	46	48	50
52	54	56	58	60
62	64	66	68	70
72	74	76	78	80
82	84	86	88	90
92	94	96	98	100

Hypothesis 2:
Multiples of 4

	4		8	
12		16		20
	24		28	
32		36		40
	44		48	
52		56		60
	64		68	
72		76		80
	84		88	
92		96		100

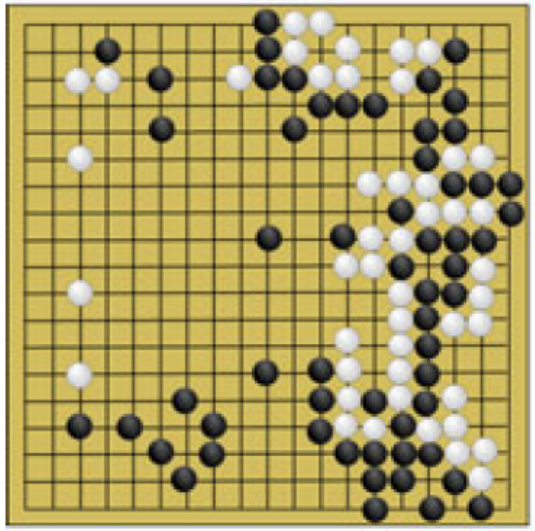
Hypothesis 3:
Powers to the 2

1	4			9
		16		
		25		
		36		
				49
	64			
81				
				100

There are other tasks in unsupervised learning, such as clustering and dimensionality reduction.

Reinforcement learning is when the machine, known as agent, gets rewards or punishments according to a given rule for its decisions, so that the agent learns an optimal strategy by trial and error.

Example 4: Board games



Deep learning is a technique to perform machine learning inspired by our brain's own network of neurons.

Ingredients of machine learning

Definition: Supervised learning task

Given an input domain $\mathcal{X}$ and an output domain $\mathcal{Y}$, a dataset $\mathcal{D} = \{(x_1, y_1), \ldots$ of pairs $(x_m, y_m) \in \mathcal{X} \times \mathcal{Y}$, $m = 1, \ldots, M$ that contain input x_m and target outputs y_m sampled from a probability distribution $p(x, y)$, as well as a new unclassified input $x \in \mathcal{X}$, predict the corresponding output $y \in \mathcal{Y}$.

Definition: Unsupervised learning task

Given an input domain $\mathcal{X}$ and a dataset $\mathcal{D} = \{x_1, \ldots, x_M\}$ of input samples $x_m \in \mathcal{X}$ drawn from a probability distribution $p(x)$, draw a new sample from p.

Machine learning solves problems by defining a model family $\{f\}$, which is mathematically expressed as a set of functions f that map inputs from $\mathcal{X}$ to outputs from $\mathcal{Y}$, or of distributions p over the inputs in $\mathcal{X}$.

In supervised learning, one picks the best model to reproduce the data, formally known as *risk minimization*, by defining a loss function that measures the disparity between the model's prediction $f(x)$ for an input x and the target output y.

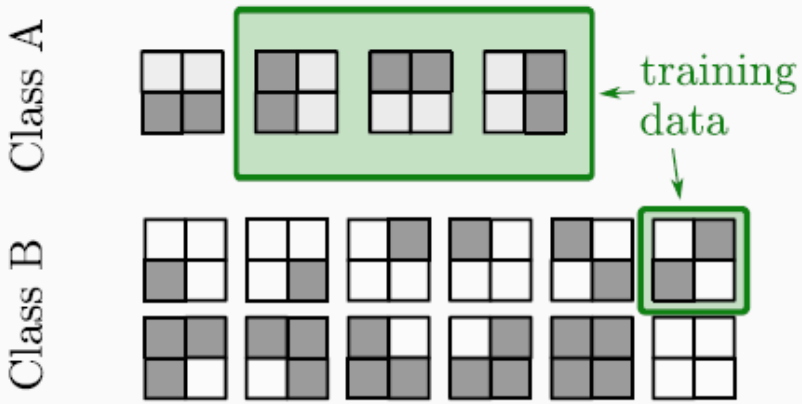
The best model is the one that minimizes the expected loss over all possible data.

The challenge is to select a model that generalizes from the limited examples to the entire domain of the data.

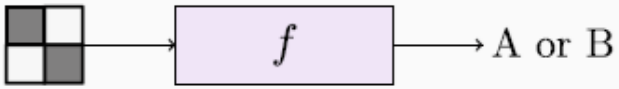
In unsupervised learning, we do not have a loss function comparing predictions and labels, but we can phrase it as the distance between the true probability distribution and the model distribution.

Machine learning models are typically parametrized. Picking the best model translates to an optimization task, in which the best parameter set θ needs to be found.

1. Data



2. Model

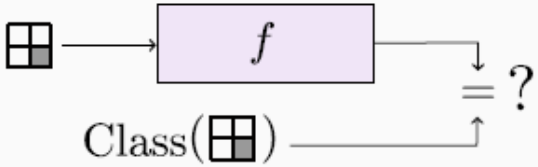


f_1: It is A if a row or column is filled, else B

f_2: It is A if two blocks are filled, else B

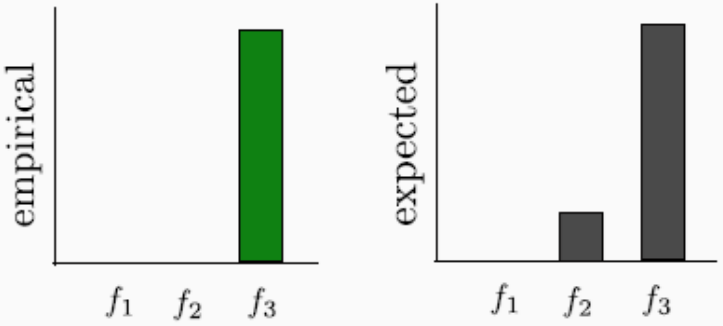
f_3: It is A if no block is filled, else B

3. Loss



$$L(f(\boxplus),\mathrm{B}) = \begin{cases} 0 \text{ if } f(\boxplus) = \mathrm{B} \\ 1 \text{ else} \end{cases}$$

4. Risk



Classification or regression

The choice of the output domain in supervised learning determines whether we are solving classification or regression tasks.

If $\mathcal{Y}$ is a set of D discrete class labels $\{l_1, \ldots, l_D\}$, then it is a classification task.

Multi-class classification task can always be reduced to binary classification.

Regression refers to problems in which the prediction y is continuous-valued but does not necessarily have a probabilistic interpretation.

A classification task can be generalized to regression by switching to continuous variables and adjusting the functions or distributions accordingly, while the outcome of regression can be discretized.

Data

The dataset is assumed to be *independent and identically distributed* (*iid*), even though in reality this may not be the case.

The data domain $\mathcal{X}$ is assumed to be the space $\mathbb{R}^N$. Such data samples are called *feature vectors*.

In cases where the raw data is not from a numerical domain, one has to first find a suitable vectorial representation.

Data is often pre-processed to improve the predictive power of an algorithm.

1. Feature selection: The choice of features of an instance to consider;
2. Feature scaling: Statistical transformation of data;
3. Feature engineering: Hand-crafting features, for example combining several features in the original dataset into a single one.

Models

Models in machine learning can be defined as functions, distributions, algorithms, or rules that reproduce the properties of a set of data samples.

Definition: Deterministic model

Let $\mathcal{X}$ be an input domain and $\mathcal{Y}$ be an output domain for a supervised machine learning problem. A deterministic model is a function

$$f : \mathcal{X} \to \mathcal{Y}, \qquad f(x) = y,\, x \in \mathcal{X}, y \in \mathcal{Y}. \tag{2}$$

As mentioned previously, a model can depend on a set of parameters θ and hence defines a model family $\{f_\theta\}$.

There may also be *hyperparameters* that give the model family additional degree of freedom.

For example, the hyperparameter for the nearest neighbour method in the previous discussion is the distance measure, i.e. the squared distance.

Definition: Probabilistic model

Let $\mathcal{X}$ be an input domain and $\mathcal{Y}$ be an output domain of a machine learning problem. A probabilistic model for supervised learning is defined by a full probability distribution

$$p_\theta : \mathcal{X} \times \mathcal{Y} \to [0, 1], \qquad p_\theta(x, y),\ x \in \mathcal{X}, y \in \mathcal{Y}, \tag{3}$$

or a conditional distribution

$$p_\theta : \mathcal{X} \times \mathcal{Y} \to [0, 1], \qquad p_\theta(y|x),\ x \in \mathcal{X}, y \in \mathcal{Y}. \tag{4}$$

An unsupervised probabilistic model is defined by the distribution

$$p_\theta : \mathcal{X} \to [0, 1], \qquad p_\theta(x),\ x \in \mathcal{X}. \tag{5}$$

A probabilistic model can work in two ways.

1. It defines an actual function that computes the probability p for a data sample. We call it *density estimator*.
2. It produces samples from the probability distribution p. We call it *generative model*. One example is the generative adversarial network, which are deterministic model that are fed with noise and produce samples from a distribution.

Probabilistic model may depend on parameters θ too, such as the mean μ and variance σ^2 from the normal distribution,

$$p_{\mu,\sigma}(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \tag{6}$$

or q from Bernoulli distribution,

$$p_q(x) = q^x(1-q)^{1-x}. \tag{7}$$

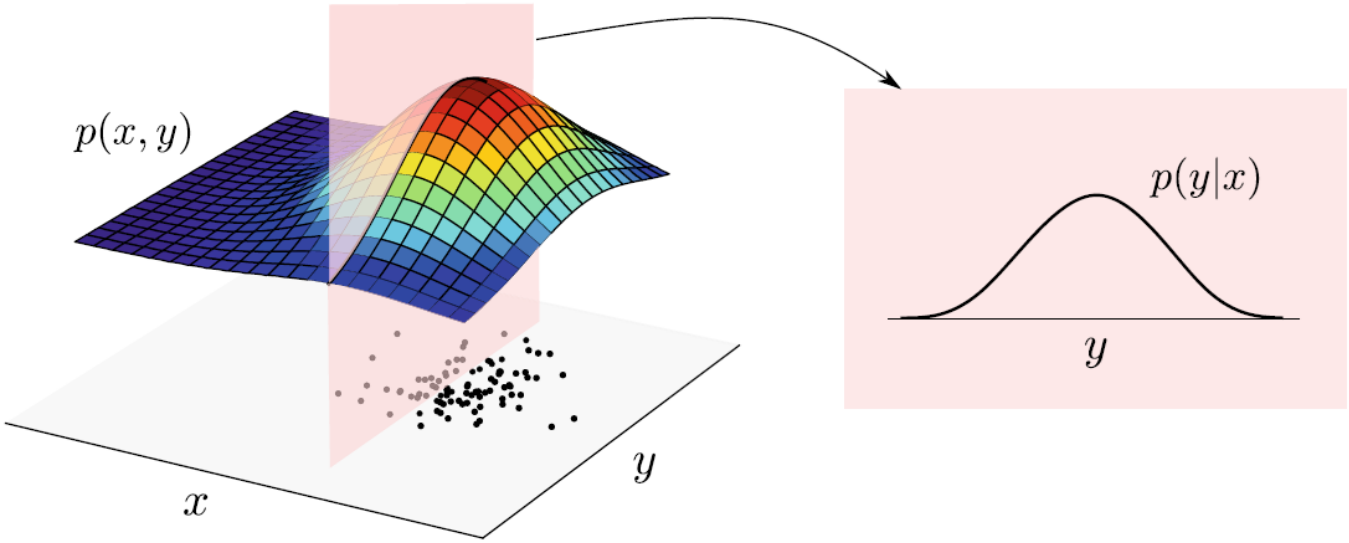
Depending on contexts, the above parameters may be identified as hyper-parameters.

A supervised probabilistic model can be used as a deterministic model if we extract the conditional distribution $p_\theta(y|x)$ and provide a rule of deriving deterministic outputs $y = f(x)$ from it, either by the *maximum a posteriori estimate*

$$f_\theta(x) = \max_y p_\theta(y|x), \tag{8}$$

or by the mean of the distribution

$$f_\theta(x) = \int p_\theta(y|x) y dy. \tag{9}$$

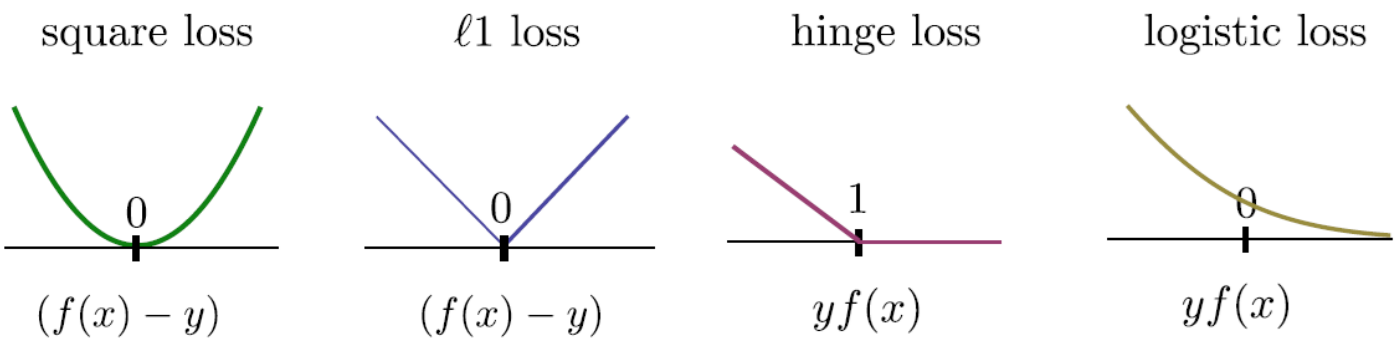


Loss function

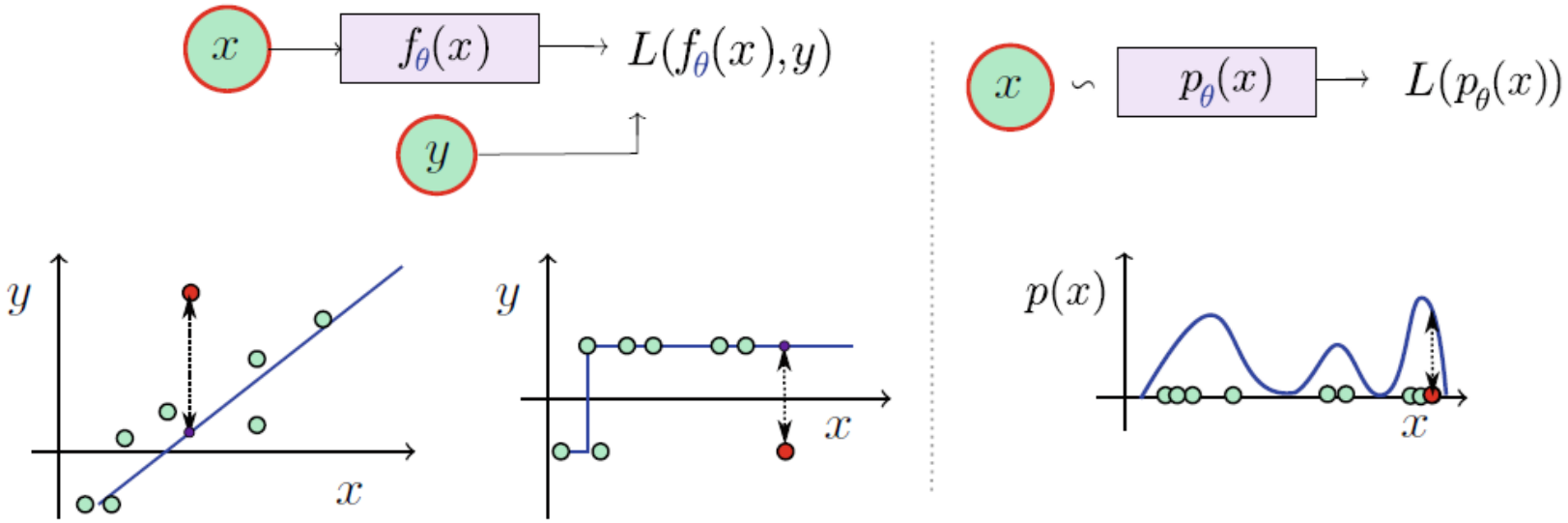
Loss is a measure of how close the model predictions are to the target labels.

For supervised learning, there are many ways to define loss function:

1. Accuracy: $\text{accuracy} = \frac{\text{Number of correctly classified examples}}{\text{Total number of examples}}$
2. Mean-square-error loss: $L(f_\theta(x), y) = (f_\theta(x) - y)^2$
3. ℓ_1 loss: $L(f_\theta(x), y) = |f_\theta(x) - y|$
4. Hinge loss: $L(f_\theta(x), y) = \max(0, 1 - yf_\theta(x))$
5. Logistic loss: $L(f_\theta(x), y) = \log(1 + e^{-yf_\theta(x)})$



In unsupervised learning, the quality of a model is harder to define.

If the model provides us with the probability $p(x)$, we may introduce a likelihood $L(x) = p_\theta(x)$ to measure how likely the observed data was sampled from the model p_θ.

If the model only returns samples x, we may turn unsupervised into supervised learning, where the labels are whether a sample came from the model or the original distribution, which allows us to use supervised loss functions.